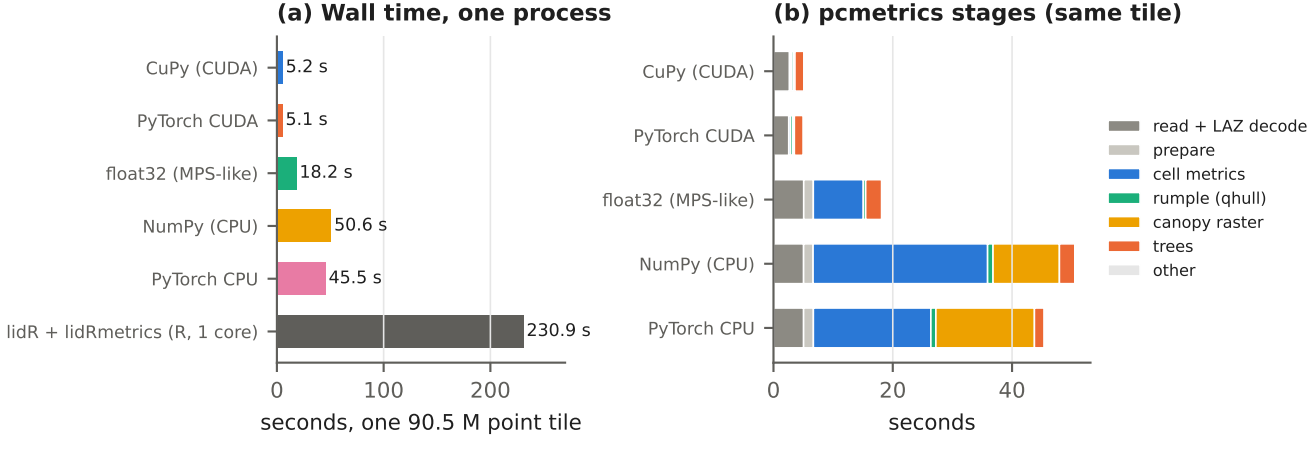
(a) Wall time, one process
CuPy (CUDA)
5.2 s
PyTorch CUDA
5.1 s
float32 (MPS-like)
18.2 s
NumPy (CPU)
50.6 s
PyTorch CPU
45.5 s
lidR + lidRmetrics (R, 1 core)
230.9 s
0
100
200
seconds, one 90.5 M point tile
(b) pcmetrics stages (same tile)
CuPy (CUDA)
PyTorch CUDA
float32 (MPS-like)
NumPy (CPU)
PyTorch CPU
0
20
40
seconds
read + LAZ decode
prepare
cell metrics
rumple (qhull)
canopy raster
trees
other